Math 105L **Lab 3 Part 1**

Exploring Limits: Spreadsheets, Indeterminate Forms, & Holes

1. Numerical Limits via Spreadsheets

In this laboratory, students explore the behavior of functions near points of interest by constructing numerical tables that approach a target value $x = c$ from both directions:

- **Approaching from the Right ($x > c$):** Step sizes $x = c + 0.1, c + 0.01, c + 0.001, c + 0.0001$.
- **Approaching from the Left ($x < c$):** Step sizes $x = c - 0.1, c - 0.01, c - 0.001, c - 0.0001$.
- **Convergence Criterion:** If values from both sides converge to the identical number L, we deduce numerically that $\lim_{\mathbf{x \to c}} \mathbf{f(x) = L}$.

Indeterminate Forms: The Meaning of $\frac{0}{0}$ When direct evaluation yields $\frac{0}{0}$, this does **not** mean the limit is undefined or impossible! Rather:

- $\frac{0}{0}$ is an **indeterminate form**, signaling that both numerator and denominator share a common factor $(x - c)$ that vanishes at $x = c$.
- Algebraic factoring allows cancellation of the common factor for all $x \neq c$, revealing the true limiting behavior.

2. Classroom Demonstration on Blackboard: $f(x) = \frac{2x^2 - x - 1}{x - 1}$ as $x \to 1$

- **Step 1: Attempt Direct Substitution:**

$$f(1) = \frac{2(1)^2 - (1) - 1}{1 - 1} = \frac{2 - 2}{0} = \mathbf{\frac{0}{0}} \quad \text{(Indeterminate!)}$$

- **Step 2: Factor the Numerator:** Find two binomials whose product is $2x^2 - x - 1$:

$$2x^2 - x - 1 = \mathbf{(2x + 1)(x - 1)}$$

- **Step 3: Cancel Common Non-Zero Factors ($x \neq 1$):**

$$f(x) = \frac{(2x + 1)(x - 1)}{x - 1} = \mathbf{2x + 1} \quad \text{for all } x \neq 1$$

- **Step 4: Compute the Limit:**

$$\lim_{x \to 1} f(x) = \lim_{x \to 1} (2x + 1) = 2(1) + 1 = \mathbf{3} \qquad \checkmark$$

3. Geometric Interpretation of Removable Discontinuities

- **Graph Comparison:** The graph of $f(x) = \frac{2x^2 - x - 1}{x - 1}$ is identical in every respect to the continuous straight line $y = 2x + 1$, with slope $m = 2$ and y-intercept $(0, 1)$.
- **The Removable Hole:** At the single point $x = 1$, the function has an **open circle (hole)** at $(1, 3)$.
- **Limit vs. Value:** The value $f(1)$ does not exist (undefined), yet the limit $\lim_{x \to 1} f(x)$ exists and equals 3.